

L'ETAPE, QC, Canada

WMO: 713820

Lat: 47.5622N

Lon: 71.2289W

Elev: 2596

StdP: 13.37

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.2	-22.1	-34.5	0.9	-27.0	-28.7	1.2	-22.0	16.9	17.8	15.6	14.0	3.0	10	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	76.3	62.5	73.2	60.7	70.4	59.7	66.2	71.3	64.3	69.3	62.6	67.5	7.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	99.7	68.4	62.6	93.4	66.8	60.7	87.4	64.9	32.4	71.5	30.9	69.5	29.5	67.5	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.6	13.1	11.7	-36.5	81.3	6.0	3.2	-40.8	83.6	-44.3	85.5	-47.7	87.3	-52.1	89.6		
			-36.7	69.2	6.1	2.2	-41.1	70.7	-44.7	72.0	-48.1	73.2	-52.6	74.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.3	3.0	5.4	15.0	28.9	42.9	53.2	58.2	56.1	49.2	37.2	24.4	11.7
	DBStd	21.58	13.73	12.76	12.53	9.35	7.39	6.97	5.30	5.53	7.36	8.29	9.89	12.09
	HDD50	7185	1457	1248	1085	633	243	43	4	11	102	402	769	1188
	HDD65	11971	1922	1668	1550	1082	686	357	220	279	474	861	1219	1653
	CDD50	706	0	0	0	0	22	140	257	201	79	7	0	0
	CDD65	18	0	0	0	0	0	4	9	4	1	0	0	0
	CDH74	205	0	0	0	0	8	76	75	33	13	0	0	0
	CDH80	14	0	0	0	0	0	8	5	0	1	0	0	0
Wind	WSAvg	5.2	5.1	5.2	5.7	5.7	5.1	5.0	4.9	4.7	5.2	5.4	5.6	5.2
Precipitation	PrecAvg	51.2	3.0	2.9	3.8	3.4	4.1	4.6	5.6	4.4	5.2	4.8	4.5	4.4
	PrecMax	61.1	5.2	6.7	6.0	6.5	6.7	7.9	8.3	10.2	10.8	9.1	6.8	7.9
	PrecMin	43.5	0.9	1.4	1.8	1.7	1.6	2.6	1.7	1.9	2.0	1.8	1.3	1.5
	PrecStd	4.8	1.3	1.3	1.3	1.3	1.5	1.4	1.7	1.9	1.9	1.6	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	38.7	48.6	59.1	74.2	80.7	80.2	77.6	75.5	64.8	52.7	41.0
		MCWB	41.6	36.2	39.4	46.9	56.2	64.1	64.7	63.8	63.4	54.8	49.6	40.5
	2%	DB	34.1	33.9	41.3	52.3	68.6	75.8	75.8	74.2	69.7	58.9	46.2	35.0
		MCWB	33.7	32.3	35.9	43.2	52.5	60.9	63.1	62.0	60.9	53.9	44.0	34.4
	5%	DB	28.4	29.4	37.3	47.5	63.3	71.7	72.6	70.8	66.1	54.4	41.8	32.1
		MCWB	27.8	27.8	33.1	39.7	51.3	58.8	61.4	60.8	59.5	49.7	39.8	31.7
	10%	DB	23.3	25.7	33.7	42.9	58.4	67.7	69.9	67.9	62.4	50.3	37.3	28.5
		MCWB	22.5	24.2	31.1	36.8	49.3	57.2	60.7	59.5	57.4	47.3	35.9	27.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	37.6	43.7	50.5	61.5	67.4	69.1	67.2	66.5	58.0	51.2	40.4
		MCDB	42.0	38.1	45.5	55.2	68.1	75.9	75.1	72.9	71.4	60.1	51.7	40.9
	2%	WB	33.7	32.5	37.5	45.7	57.2	64.4	66.6	64.9	63.7	54.4	44.5	34.3
		MCDB	33.9	33.6	40.0	50.2	62.5	71.4	71.0	69.7	67.2	58.4	45.4	34.5
	5%	WB	27.7	27.8	33.8	40.7	53.3	62.0	64.5	63.1	61.0	50.8	40.1	31.3
		MCDB	28.3	29.1	36.4	45.4	58.7	67.8	69.3	67.7	64.6	53.5	41.4	31.8
	10%	WB	22.6	24.3	31.1	37.6	50.4	59.5	62.5	61.5	58.3	47.4	36.0	27.5
		MCDB	23.3	25.7	33.9	42.0	57.5	65.2	67.5	66.0	61.7	49.8	37.3	28.1
Mean Daily Temperature Range	5% DB	MDBR	20.9	22.3	22.6	19.1	21.2	21.4	19.1	19.2	18.1	14.1	13.1	16.4
		MCDBR	21.4	22.2	25.3	25.7	32.1	28.6	24.6	24.6	23.7	22.9	18.3	16.1
		MCWBR	21.2	19.9	19.9	16.9	18.1	15.5	13.6	14.1	15.0	15.6	15.6	16.5
	5% WB	MCDBR	20.7	19.8	20.4	21.1	24.0	22.3	18.8	18.7	19.5	19.3	16.6	16.5
		MCWBR	20.7	18.5	17.0	15.7	16.7	14.0	12.1	12.1	14.3	15.1	15.7	16.8
Clear-Sky Solar Irradiance	taub	0.242	0.255	0.287	0.319	0.329	0.349	0.364	0.346	0.333	0.307	0.283	0.251	
	taud	2.333	2.289	2.238	2.334	2.484	2.432	2.390	2.447	2.484	2.545	2.490	2.363	
	Ebn at Noon	273	292	297	296	296	290	283	285	279	271	255	254	
	Edh at Noon	22	29	36	37	33	35	36	33	29	23	20	19	
All-Sky Solar Radiation	RadAvg	432	723	1091	1409	1606	1720	1659	1442	1120	659	375	294	
	RadStd	55	88	86	115	167	116	176	95	87	56	41	27	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page